Kounianhua Du

Shanghai Jiao Tong University

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Education

Shanghai Jiao Tong University
Bachelor in Computer Science *Sept. 2017 - Jun. 2021*

- Cumulative GPA: 90.33, 3.91

Shanghai Jiao Tong University
Master in Computer Science *Sept. 2021 - Jun. 2023*

- Advisors: Prof. Weinan Zhang, and Prof. Yong Yu APEX LAB

Shanghai Jiao Tong University
Phd in Computer Science *Apr. 2024 - Now*

- Advisors: Prof. Weinan Zhang, and Prof. Muhan Zhang APEX LAB

Skills

- Python, C++, Matlab, Pytorch, DGL
- Cet 6: 619

Honor

- Huatai Scholarship, Weichuang Scholarship, School B Level Scholarship, Outstanding Graduate.
- Huawei Spark Honor

Internship

Tencent Youtu Lab **Shanghai**
Qingyun Internship *Feb. 2026 - Now*

- Research on cache-enhanced on policy distillation. CacheOPD (In Progress)
  - Teacher cache guided rollout for beneficial token hit rate increase.
  - Information-maintained token importance.
  - Distribution-aware curriculum training and regularization.
- Research on multi-agent skill evolution for generalizable artifacts. MultiEvoSkills
  - Structure-aware multi-agent skill evolution pipeline.
  - Distributed workload so that budget is reduced.
  - Collective intelligence contributes to strong and generalizable skill artifacts.
- Research on MCP-native agent data synthesis, reasoning-tool seesaw, reward design, and disentanglement. MCPDisCo, (ModelWeight)
  - Position MCP as the substrate of the agent net. First work of RL-tuning on MCP-Native Agents.
  - Disentangled reward with mask strategy, along with reasoning-focused gradient surgery for tool-reasoning seesaw mitigation and long-horizon task planning improvement.
  - Consistent performance gain and OOD validation.

Xiaohongshu **Shanghai**
Red Star *Jun. 2025 - Feb. 2026*

- Research on inference-time personalization for LLMs, including its applications in content generation and web function calling. Fints, FleetPilot
  - Propose an inference-time personalization framework that delivers robust personalization by treating it as a sample-level activation shift rather than a parametric update.
  - Fints shows distinct superiority under high data sparsity (*<10-shot*) and fast interest drift, while bringing low memory overhead and adding light latency.
  - Systematic evaluation on scenarios including short content generation, long content generation, and web function calling. Various ablation studies to validate Fints's flexibility, distributional robustness, superior data efficiency, and light overhead.
- Participated in research on looped data synthesis for tool calling. LoopTool
  - A model-aware data synthesis framework that iteratively diagnoses model capabilities, corrects and synthesizes training data based on failure cases.
  - Sota performance on BFCL-v3 and AceBench among 8B models.

Huawei **Shanghai**

Research Intern, Student Leader of the CodeAgent Program *Apr. 2024 - Mar. 2025*

- Research on System2-to-System1 methods. BDC
  - Novel **MC-Tree-of-Agents** algorithm, which efficiently and effectively explores the insightful System2 knowledge via mutual verification and boosting of different LLMs, armed with reward-guided pruning and refinement to prioritize high-quality reasoning paths.
  - Novel **DisenLoRA** algorithm that offers customized problem solver for robust code generation, which contains an input-aware hypernetwork to aggregate over the disentangled composable LoRA experts.
- Research on graphical view of codes, used in RAG and soft-prompting for enhanced generation. CodeGRAG
  - Effective graphical view to inform and stimulate the structural programming knowledge of LLMs.
  - Effective soft prompting technique to preserve the programming domain knowledge and inject it into LLMs parameters.
- Participate in RethinkMCTS, RetrievalPRM, DebateCoder, CodeAPEX, and NLDebugging.
- Got **over 2%** llm performance gain via model merging.

Huawei **Shanghai**

Research Intern, Participate in LLM-enhanced Recommendation Program *Mar. 2023 - Oct. 2023*

- Research on building adaptive parametric knowledge base. AdaPK
  - Adaptively feed instance-wise information need with essential and disentangled parametric knowledge compressed from voluminous data.
  - Propose two rules to regularize parametric knowledge base for robustness and generalization without increasing the size of it.
- Research on collaborating tabular and semantic space for prediction. DisCo
  - Propose to capture both specificity and consistency from tabular and semantic space with the dual-side attentive network under the regularization of the proposed sufficiency and disentanglement constraints.
  - Emphasize the importance of unique and disentangled information, and propose the first work to disentangle the tabular and semantic representation spaces for unique domain knowledge.
- Participate in Rella, SinKT, ElcoRec.

Amazon **Shanghai**

Applied Scientist, Software Engineer *Nov. 2020 - Feb. 2023*

- Research on hypergraph applications for tabular data prediction. PET
  - Propose a **retrieval-based** hypergraph model for tabular data prediction, where relevant samples distribution is obtained for robust prediction.
  - Unify the feature enhancement and label propagation to obtain robust representations.
- Participated in researches on graph-based interaction models and logical GNN.
  - GraphANGEL, GraphHINGE
- Developed DGL-based models. Repo
  - Including GNN-FiLM, NGCF, TAHIN, GNNExplainer, FATE, Grape, TabGNN, HIRS, etc.

Researches

- **LLM Agents**

- **Kounianhua Du**, Ruoshui Shan, Jianxing Liu, Yucheng Zhang, Ke Li, Xing Sun, Weinan Zhang. MultiEvoSkills: Structure-Aware Multi-Agent Evolution for Generalizable Skill Artifacts. In submission.
- **Kounianhua Du**, Jianxing Liu, Shuai Shao, Ke Li, Xing Sun, Weinan Zhang. MCPDisCo: Seesaw-Mitigated Tuning for MCP-Native Agents. In submission.
- **Kounianhua Du**, Jianxing Liu, Kangning Zhang, Wenxiang Jiao, Yuan Lu, Jiarui Jin, Jianghao Lin, Weiwen Liu, Yong Yu, Weinan Zhang. FleetPilot: Fine-Grained Steering Empowers Real-World Personalization.
- **Kounianhua Du**, Jianxing Liu, Kangning Zhang, Wenxiang Jiao, Yuan Lu, Jiarui Jin, Weiwen Liu, Yong Yu, Weinan Zhang. Fints: Efficient Inference-Time Personalization for LLMs with Fine-Grained Instance-Tailored Steering.
- Kangning Zhang, Weiwen Liu, Wenxiang Jiao, **Kounianhua Du**, Yuan Lu, Weinan Zhang, Lei Zhang, Yong Yu. LoopTool. Closing the Data–Training Loop for Robust LLM Tool Calls. **ACL Main 2026**.
- **Kounianhua Du**, Hanjing Wang, Jianxing Liu, Jizheng Chen, Xinyi Dai, Yasheng Wang, Ruiming Tang, Yong Yu, Jun Wang, Weinan Zhang. Boost, Disentangle, and Customize: A Robust System2-to-System1 Pipeline for Code Generation. **ACL Finding 2025**.
- Jizheng Chen, **Kounianhua Du**, Xinyi Dai, Weiming Zhang, Xihuai Wang, Yasheng Wang, Ruiming Tang, Weinan Zhang, Yong Yu. DebateCoder: Towards Collective Intelligence of LLMs via Test Case Driven LLM Debate for Code Generation. **ACL Main 2025.**.
- Jiachen Zhu, Congmin Zheng, Jianghao Lin, **Kounianhua Du**, Ying Wen, Yong Yu, Jun Wang, Weinan Zhang. Retrieval-Augmented Process Reward Model for Generalizable Mathematical Reasoning. **ACL Finding 2025**.
- Qingyao Li, Wei Xia, **Kounianhua Du**, Xinyi Dai, Ruiming Tang, Yasheng Wang, Yong Yu, Weinan Zhang. RethinkMCTS: Refining Erroneous Thoughts in Monte Carlo Tree Search for Code Generation. **EMNLP 2025**.
- Weiming Zhang, Qingyao Li, Xinyi Dai, Jizheng Chen, **Kounianhua Du**, Weiwen Liu, Yasheng Wang, Ruiming Tang, Yong Yu, Weinan Zhang. NL-Debugging: Exploiting Natural Language as an Intermediate Representation for Code Debugging. **EMNLP 2025**.
- Ning Li, Xiangmou Qu, Jiamu Zhou, Jun Wang, Muning Wen, **Kounianhua Du**, Xingyu Lou, Qiuying Peng, Weinan Zhang. MobileUse: A GUI Agent with Hierarchical Reflection for Autonomous Mobile Operation. **Neurips 2025**.
- **Kounianhua Du**, Jizheng Chen, Renting Rui, Huacan Chai, Wei Xia, Yasheng Wang, Ruiming Tang, Yong Yu, Weinan Zhang. CodeGRAG: Bridging the Gap between Natural Language and Programming Language via Graphical Retrieval Augmented Generation. **Huawei Spark Honor (2024)**.
- Weinan Zhang, Junwei Liao, Ning Li, **Kounianhua Du**, Jianghao Lin. Agentic Information Retrieval.
- Lingyue Fu, Huacan Chai, Shuang Luo, **Kounianhua Du**, Weiming Zhang, Longteng Fan, Jiayi Lei, Renting Rui, Jianghao Lin, Yuchen Fang, Yifan Liu, Jingkuan Wang, Siyuan Qi, Kangning Zhang, Weinan Zhang, Yong Yu. Codeapex: A bilingual programming evaluation benchmark for large language models.

Recommendation

- **Kounianhua Du**, Jizheng Chen, Jianghao Lin, Hangyu Wang, Xinyi Dai, Bo Chen, Ruiming Tang, Weinan Zhang. DisCo: Towards Harmonious Disentanglement and Collaboration between Tabular and Semantic Space for Recommendation. **KDD 2024**.
- Jizheng Chen*, **Kounianhua Du***, Jianghao Lin, Bo Chen, Ruiming Tang, Weinan Zhang. ELCoRec: Enhance Language Understanding with Co-Propagation of Numerical and Categorical Features for Recommendation. **CIKM 2024**.

- Lingyue Fu, Hao Guan, **Kounianhua Du**, Jianghao Lin, Wei Xia, Weinan Zhang, Ruiming Tang, Yasheng Wang, Yong Yu. Sinkt: A structure-aware inductive knowledge tracing model with large language model. **CIKM 2024**.
- **Kounianhua Du**, Jizheng Chen, Jianghao Lin, Bo Chen, Menghui Zhu, Ruiming Tang, Weinan Zhang and Yong Yu. Extracting Essential and Disentangled Knowledge for Recommendation Enhancement. **DLP 2023**.
- Jianghao Lin, Rong Shan, Chenxu Zhu, **Kounianhua Du**, Bo Chen, Shigang Quan, Ruiming Tang, Yong Yu, Weinan Zhang. ReLLa: Retrieval-enhanced Large Language Models for Lifelong Sequential Behavior Comprehension in Recommendation. **WWW 2024**.

Graph Data Mining

- **Kounianhua Du**, Weinan Zhang, Ruiwen Zhou, Yangkun Wang, Xilong Zhao, Jiarui Jin, Quan Gan, Zheng Zhang, David P Wipf. Learning enhanced representation for tabular data via neighborhood propagation. **Neurips 2022**.
- Jiarui Jin, Yangkun Wang, **Kounianhua Du**, Weinan Zhang, Quan Gan, Zheng Zhang, Yong Yu, David Wipf. Inductive Relation Prediction Using Analogy Subgraph Embeddings . **ICLR 2022**.
- Jiarui Jin, **Kounianhua Du**, Weinan Zhang, Jiarui Qin, Yuchen Fang, Yong Yu, Zheng Zhang, Alexander J. Smola. Learning Interaction Models of Structured Neighborhood on Heterogeneous Information Network. **TOIS 2021**.
- Jiarui Jin, Jiarui Qin, Yuchen Fang, **Kounianhua Du**, Weinan Zhang, Yong Yu, Zheng Zhang, Alexander J. Smola. An Efficient Neighborhood-based Interaction Model for Recommendation on Heterogeneous Graph. **KDD 2020**.